

Sherlock Holmes and the Case of the **Green Dashboard**

*Everything was green. The trace was missing. **The infrastructure knew.***



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— DRAMATIS PERSONAE

The cast

SHERLOCK HOLMES

Me, this afternoon

Your consulting detective for the hour. My method is observability: infer the internal state of a system from its outputs, and always ask why.

DR. WATSON

You, the audience

The on-call engineer. Paged at 03:07 with a red pager in one hand and a green wall of dashboards in the other. This afternoon you learn the method.

INSPECTOR LESTRADE

Monitoring

Honest, diligent, limited. Raises an alarm when a metric chosen in advance crosses a threshold set in advance.

MRS. HUDSON

The infrastructure

Keeps the house running. Sees everything that happens downstairs. Nobody ever thinks to ask her.

PROF. MORIARTY

The culprit

Hides inside a complex system, where so many variables interact that it feels unknowable. You meet him in chapter VIII.



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— ACT ONE · A STUDY IN GREEN

Everything was green.

A ringing phone and a green wall

At 03:07 the support line lit up, Watson. Merchants across the country reported failed payments. No alarms were given. You opened the wall of dashboards and every panel glowed a reassuring green.

03:07

Downtime costs about \$9000 a minute. Not one alarm had fired.

"Come, Watson, come! The game is afoot." — The Abbey Grange

the telephone · the wall

caller	support desk
reports	41 merchants in 6 minutes
alarms	0 firing
dashboard	ALL GREEN
health	200 OK
processor	31 % memory 54 %

The question Lestrade never asked

Lestrade filed his report at 03:09, Watson. Every check had passed and every number sat inside its limit. 'Nothing to see here,' he said, and he meant it. Read the fifth line again.

INSPECTOR LESTRADE · INCIDENT REPORT · 03:09

```
PASS health endpoint      200 OK
PASS processor            31 %      limit 80 %
PASS memory               54 %      limit 85 %
PASS error rate           0.4 %     limit 1 %
PASS database connections 98 / 100 limit 100
conclusion                 no anomaly · alarms raised: 0
```

HOLMES · WHAT LESTRADE COULD NOT ASK

Monitoring answers the questions written down in advance, against limits set in advance. It saw.

Observability infers the inside of the system from its outputs and asks the question nobody wrote down. It observes.

Ninety-eight connections out of a hundred is a pass, Watson. It is also a queue.

"You see, but you do not observe." — A Scandal in Bohemia

Five kinds of telemetry, five questions

Empty the bag onto the table, Watson: the ledger, the diary, the footprints, the autopsy, the calendar. Each witness answers a different question. Ask the wrong one and it answers honestly, and uselessly.

METRICS · THE LEDGER

Is something wrong?

Numbers at a point in time, enriched with labels. Cheap to store, fast to query.

LOGS · THE DIARY

What happened?

Structured or not, huge volume, slow to search unless you filter first.

TRACES · THE FOOTPRINTS

Where did it go?

trace_id for the whole journey, span_id for every step along it.

PROFILES · THE AUTOPSY

Why is it slow?

Processor cycles and memory, function by function. Continuous, not sampled once.

EVENTS · THE CALENDAR

What changed?

Deploys, config changes, CI/CD (continuous integration and delivery). What happened five minutes before the alert.

"It is a capital mistake to theorize before one has data." — A Scandal in Bohemia

The laboratory

Every witness gives a statement, Watson. A constable collects them, each statement goes into its cabinet, and I read them all at one desk. That is the laboratory.

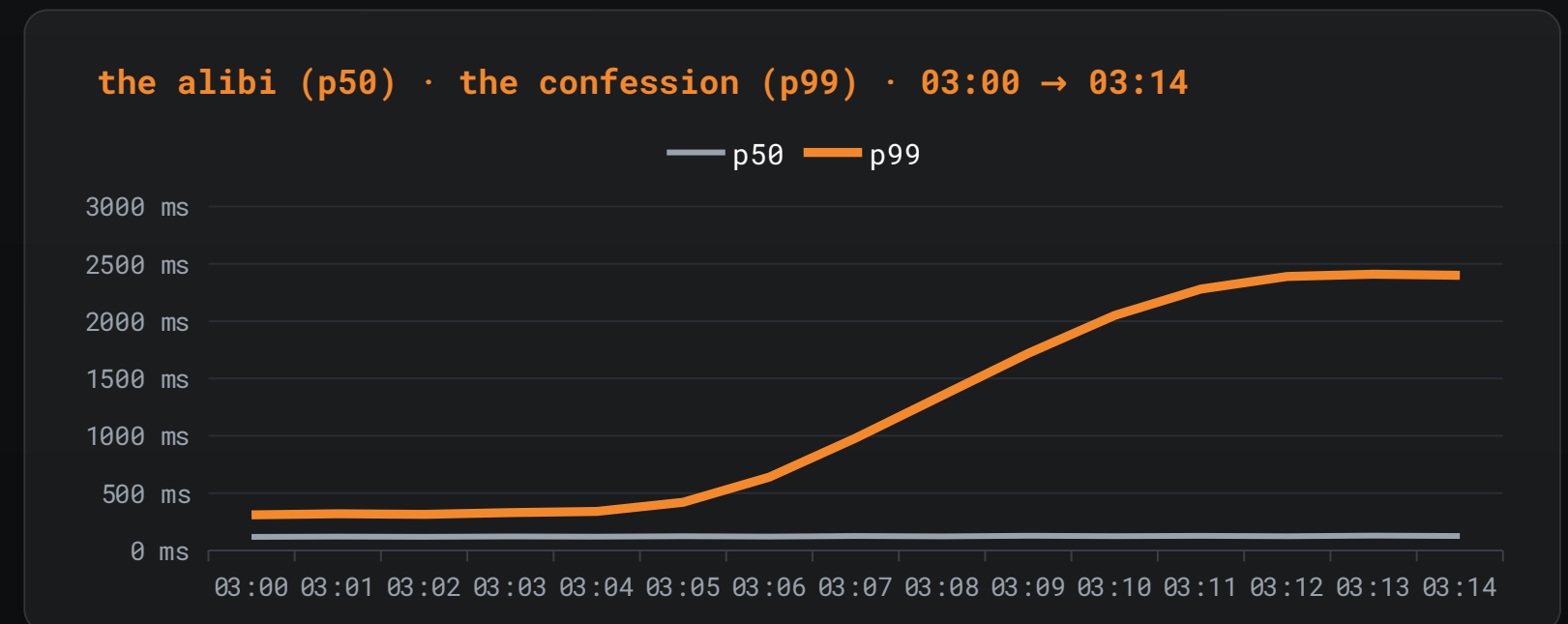


"Data! Data! Data! I can't make bricks without clay." — The Copper Beeches

The average had an alibi

You showed me the latency panel, Watson: a flat, comfortable line. I asked for the ninety-ninth percentile, and it climbed like a man running up the stairs.

- Golden signals: latency, traffic, errors, saturation. Lestrade watched one of them, averaged. That average is the alibi.
- RED (rate, errors, duration) for services, USE (utilisation, saturation, errors) for resources.

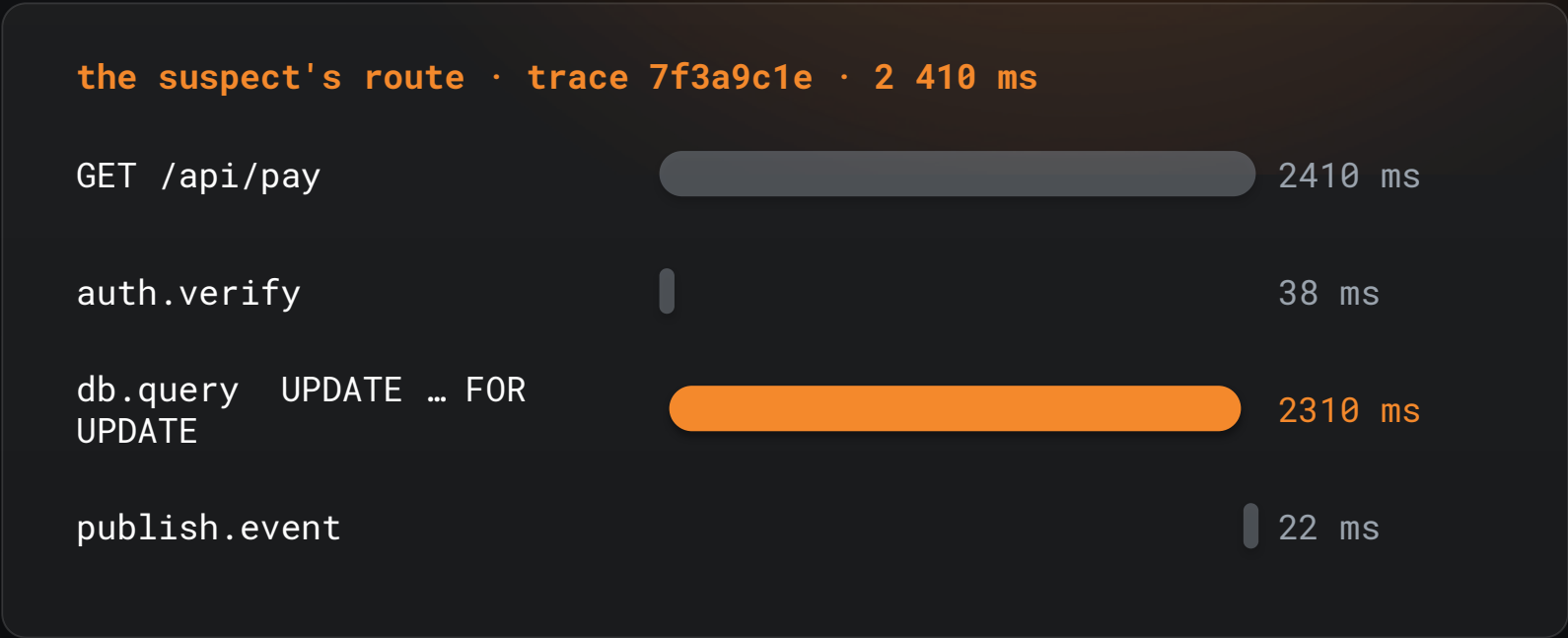


"The world is full of obvious things which nobody by any chance ever observes." — The Hound of the Baskervilles

From the crowd to a single suspect

A metric is a crowd, Watson. We need one man out of it. Touch a single point on the p99 line and the crowd parts: one request, one trace, one journey.

- Exemplars pivot from the aggregated view of a metric to one request in a trace.
- Every span records span_id, parent_id, start and end. Together they are the journey.



"Singularity is almost invariably a clue." — The Boscombe Valley Mystery

Loki, and the art of asking

The logs had seen everything and said nothing, Watson, because nobody asked them properly. Take the trace ID from the slow span and ask Loki one precise question.

- Loki indexes labels only. Lines live in compressed chunks in object storage. Filter first.
- Good labels: namespace, cluster, job, app, instance. Poor: userid, traceid, path, status code.

```
# the interrogation, in LogQL
{app="checkout", env="prod"}
  |= "7f3a9c1e"           # trace_id lives in the line
  | logfmt | level="error"

03:07:12 msg="lock wait timeout"
table=wallet_ledger waited=2310ms
blocked_by=pid 4471 (reconcile-daily)
```

"There is nothing more deceptive than an obvious fact." — *The Boscombe Valley Mystery*

Moriarty was in the basement all along.

The log named a process. The trace stopped at the database door. So I went downstairs, Watson, to the one witness nobody had interviewed, Mrs. Hudson, the infrastructure. She had been watching him for three hours.

THE CULPRIT · PROFESSOR MORIARTY

The 06:00 reconciliation job, started at 03:00.

A cron change at 02:58, deployed with no event recorded. The job took row locks on the wallet ledger one batch at a time and held each for 2.3 seconds. Every payment queued behind it. Processor 31 %. Memory 54 %. Health check 200 OK. Everything green.

what Mrs. Hudson saw

```
crontab edited      02:58    no event recorded
pg_locks waiting   0 → 38
connections        98 / 100  (Lestrade's pass)
disk await         240 ms

sources node_exporter · postgres_exporter · syslog
```

“The little things are infinitely the most important.” — A Case of Identity

Six clicks from alert to culprit

'When you have eliminated the impossible, whatever remains, however improbable, must be the truth.' In our case, Watson, the truth was six clicks away.



Each hop is a link Grafana already knows how to make: data links, exemplars, trace to logs, logs to metrics.

Ninety seconds. No guessing. That is the whole difference between a notification and observability.

"I never guess. It is a shocking habit, destructive to the logical faculty." — *The Sign of the Four*

The alarm that should have come

Lestrade's alarm never rang, Watson, because no threshold was crossed. So we write a different kind of alarm, one that watches what the customers feel.

- SLI (service level indicator): keep three to five. SLO (service level objective): the target you agree to.
- Agree the roles before the incident.
- Cut noise in the observability system first, then in the channels. Afterwards, a blameless postmortem.

```
# the telegram Lestrade never sent · SLO 99.9 % · page at 14.4x  
burn  
( sum(rate(payment_requests_total{status="failed"}[1h]))  
  / sum(rate(payment_requests_total[1h])) ) > 14.4 * 0.001  
and  
( sum(rate(payment_requests_total{status="failed"}[5m]))  
  / sum(rate(payment_requests_total[5m])) ) > 14.4 * 0.001
```

"It is of the highest importance not to have useless facts elbowing out the useful ones." — A Study in Scarlet

Ten years of my own case notes

Every detective was once a constable, Watson. My own beat began with a SIEM (security information and event management) tool and a monitoring server, and the lessons arrived the hard way.

WHERE IT BEGAN

Icinga2, Wazuh

Wazuh as the SIEM for the security events, Icinga2 as the monitoring server for the checks. Alarms, not answers. Lestrade's desk.

BE HOLISTIC

Cover every system

Observability should reach every associated system where possible. A blind spot is exactly where the culprit stands.

IMPROVE GRADUALLY

Step by step

Set observability measures with gradual improvements. A big-bang platform is its own outage.

MIND THE COST

Observe the observer

The stack needs processor, memory and storage of its own. Give each team the signals relevant to them, not everything.

"Education never ends, Watson. It is a series of lessons, with the greatest for the last." — The Red Circle



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— ACT TWO · THE INVESTIGATION

The game is **afoot**.

Watch the clicks, not the panels

- 01 A two-service app: a checkout API (application programming interface) and a Postgres database.
- 02 Grafana Alloy ships metrics, logs and traces to Grafana Cloud at bronzecobra1867.grafana.net.
- 03 A burn-rate alert is armed, everything is green, and then I commit the crime live: Moriarty's job starts early.
- 04 You count the clicks from alert to culprit, Watson. I need six.

the click path · ≈ 90 s

- 01 Alert fires
- 02 Dashboard: p99 climbing
- 03 Exemplar → trace
- 04 Slow span → logs by trace ID
- 05 Log → Moriarty in the infra

— ACT THREE · THE BAKER STREET IRREGULARS

Your turn to deduce.

— THE RULES

Two rounds, one prize

ROUND ONE · KAHOOT

10 questions. 20 seconds each.

PromQL and LogQL (the Prometheus and Loki query languages), exemplars, burn rates, cardinality.
Hard enough that the room argues.

`kahoot.it` · `join code on screen`

ROUND TWO · SPOT THE CULPRIT

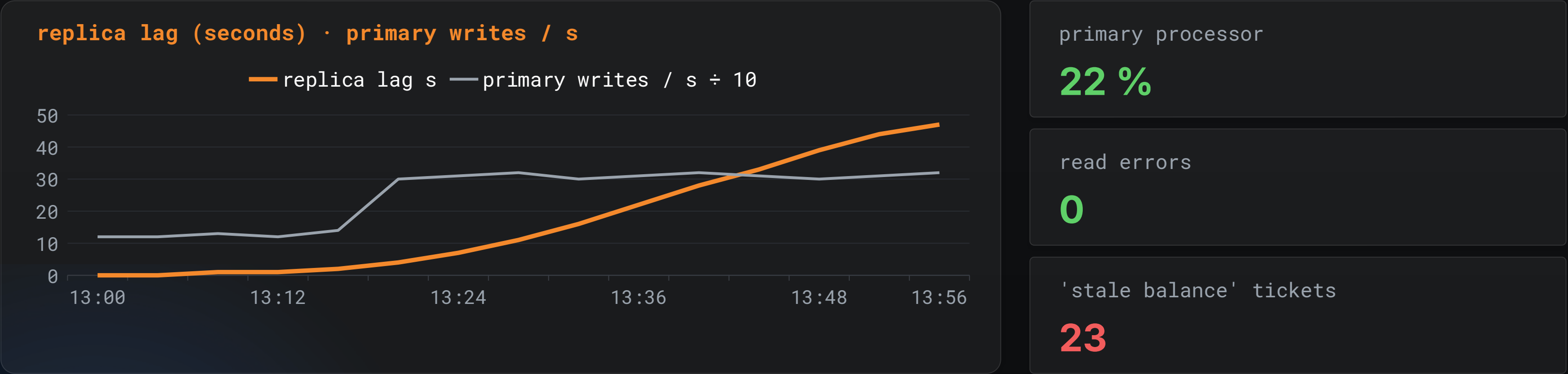
4 dashboards. 90 seconds each.

Teams of three. Name the culprit and the next witness you would question.
Replication lag, a leak, a noisy neighbour, an expired cert.

`prize: Grafana swag` · `runner-up: certificate of least trusted dashboard`

The Adventure of the Lagging Replica

Customers see yesterday's wallet balance. The primary database swears it is innocent.



90 seconds, Watson. Name the culprit, and the next witness you would question.

Elementary, Watson: replication lag

The Adventure of the Lagging Replica

THE CULPRIT

The read replica fell behind the primary. Reads still succeed, they are simply old.

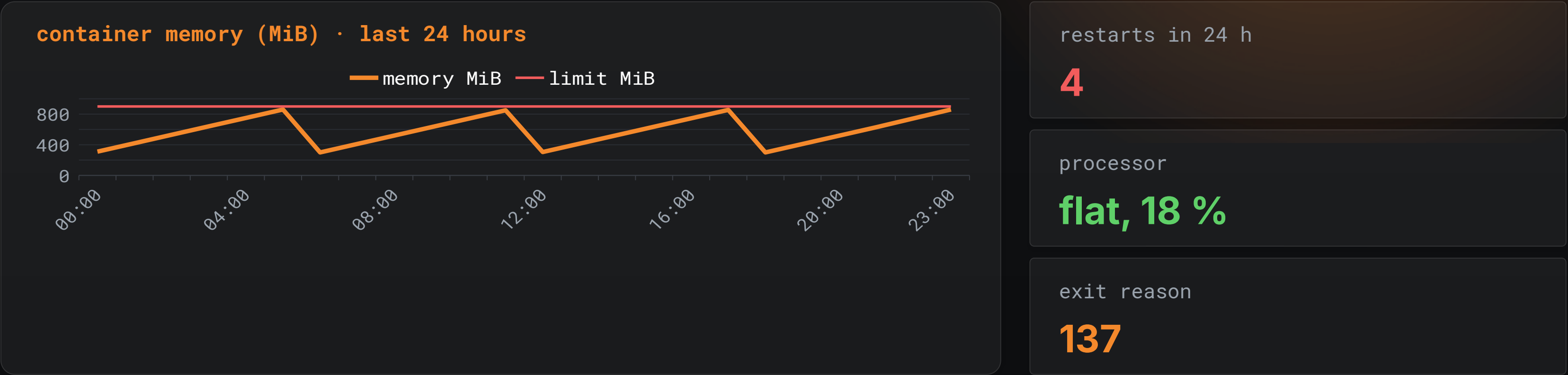
HOW HOLMES DEDUCED IT

- 01 Stale-balance tickets but zero read errors: the answers are not wrong, they are late.
- 02 Replica lag starts climbing the moment primary writes jump at 13:20, and never recovers.
- 03 Primary processor at 22 %: the primary is fine, the replica cannot apply changes fast enough.

Next witness: replication status for replay lag, replica disk I/O, then whoever started writing at 13:20.

The Case of the Swelling Heap

Every six hours the checkout service restarts itself. Nobody touched it. Everyone blames the network.



90 seconds, Watson. Name the culprit, and the next witness you would question.

Elementary, Watson: **a memory leak**

The Case of the Swelling Heap

THE CULPRIT

Memory is allocated and never released until the kernel kills the process at its limit.

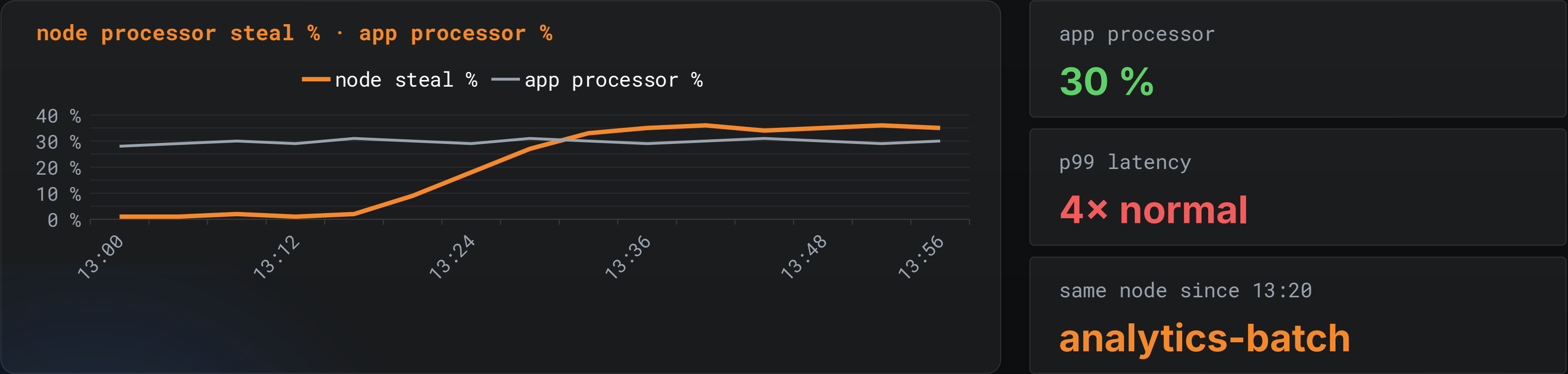
HOW HOLMES DEDUCED IT

- 01** Memory climbs in a straight line and only drops when it touches the 900 MiB limit.
- 02** Exit code 137 is 128 + 9: killed by the kernel with signal 9, the out-of-memory killer's mark.
- 03** Processor flat at 18 %: this is not load. It is memory that is never given back.

Next witness: a heap profile in Pyroscope, to find the function that keeps allocating.

The Noisy Neighbour of Flat 2B

The payment service is slow, but its own processor is barely awake. Something next door is very loud.



90 seconds, Watson. Name the culprit, and the next witness you would question.

Elementary, Watson: **a noisy neighbour**

The Noisy Neighbour of Flat 2B

THE CULPRIT

A batch pod on the same node is taking the processor time the payment service asked for.

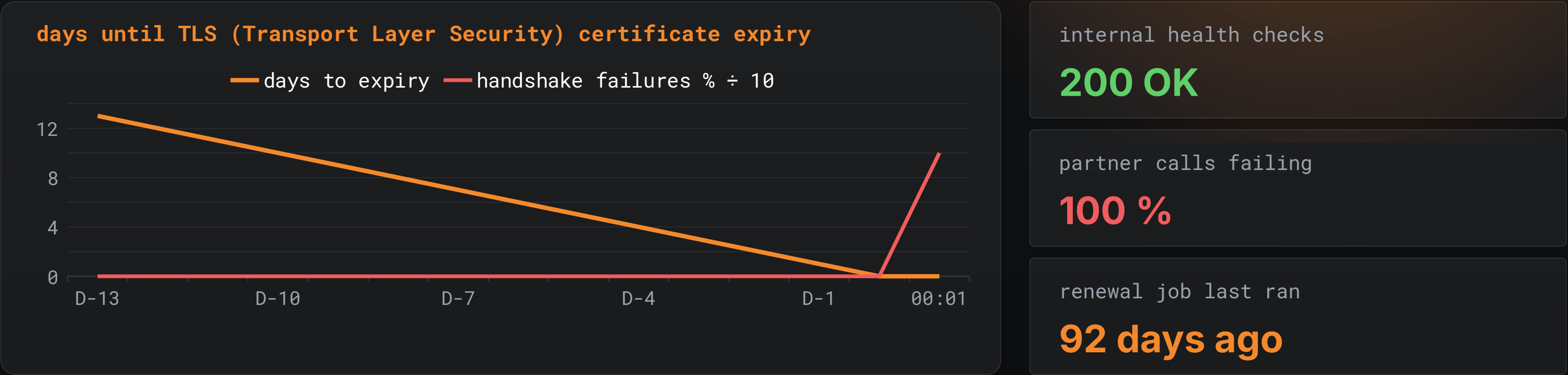
HOW HOLMES DEDUCED IT

- 01** The app's own processor stays flat at 30 %, so the app is not doing more work.
- 02** Node steal time rises from 1 % to 35 % from 13:20, exactly when analytics-batch landed on the node.
- 03** Steal is time the app wanted but the host gave away. Waiting for it is why p99 is 4× normal.

Next witness: node-level metrics and pod placement, then set requests and limits or move the batch pod.

The Adventure of the Expiring Certificate

At midnight every partner integration failed at once. Inside the building, every health check is green.



90 seconds, Watson. Name the culprit, and the next witness you would question.

Elementary, Watson: **an expired certificate**

The Adventure of the Expiring Certificate

THE CULPRIT

The TLS certificate on the partner endpoint expired at midnight, and nothing renewed it.

HOW HOLMES DEDUCED IT

- 01** Days to expiry reaches zero at midnight. Handshake failures jump to 100 % one minute later.
- 02** Internal health checks stay green because they never pass through the partner TLS endpoint.
- 03** The renewal job last ran 92 days ago, longer than a 90-day certificate lives.

Next witness: the blackbox exporter expiry probe, and an alert at 14 days to expiry, not at zero.



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— ACT FOUR · THE DEBRIEF

Questions, theories, **alibis.**

— CASE CLOSED

“You see, but you do not **observe**.”

If your alert cannot reach a trace in two clicks, it is a notification, not observability.

The infrastructure knew. Now you do too. And Moriarty is still out there, Watson.

Thank you.

Case closed. Questions?